

Lecture 22

$Ax = \lambda x$

→ Diagonalization of a matrix

$$S^{-1} A S = \Lambda$$

↓

eigenvector
matrix
↓

n-independent
eigenvectors

$$\rightarrow AS = A \begin{bmatrix} | & | & \dots \\ \lambda_1 & \lambda_2 & \dots \\ | & | & \dots \end{bmatrix} = \begin{bmatrix} | & | & \dots \\ \lambda_1 x_1 & \lambda_2 x_2 & \dots \\ | & | & \dots \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \dots & \lambda_n x_n \\ | & | & & | \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & 0 \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

↓
diagonal matrix of
eigen values.

$$AS = S \Delta \quad \text{provided } S \text{ is invertible}$$

$$\boxed{S^{-1}AS = \Delta}$$

$$\boxed{A = S \Delta S^{-1}}$$

⇒ What are eigenvectors & eigenvalues of A^2

→ If $Ax = \lambda x$

$$A^2 x = \lambda \underbrace{Ax}_{\lambda x}$$

$$\boxed{A^2 x = \lambda^2 x}$$

→ eigenvalues are squared
→ eigenvectors are same.

$$\Rightarrow A = S \Lambda S^{-1}$$

$$A^2 = (S \Lambda S^{-1})(S \Lambda S^{-1})$$

$$A^2 = S (\underbrace{\Lambda \cdot \Lambda}_{\text{diagonal matrices}}) S^{-1}$$

$$\boxed{A^2 = S \Lambda^2 S^{-1}}$$

$$\Lambda^2 = \begin{bmatrix} \lambda_1^2 & & & \\ & \lambda_2^2 & & \\ & & \ddots & \\ & & & \lambda_n^2 \end{bmatrix}$$

$$A^R = S \Lambda^R S^{-1}$$

$A^k \rightarrow 0$ as $k \rightarrow \infty$ if all

$$|\lambda_i| < 1 \quad \checkmark \checkmark$$

$\hookrightarrow$ A is sure to have n -independent eigenvectors if all the λ 's are different.

$\Rightarrow$ Repeated possibility

- may or may not have n independent eigenvectors. $\checkmark$

↳ There is a maybe here because we might still have independent eigenvectors after same eigenvalue eg.

I

$$\lambda_i = 1 \quad x = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dots$$

Triangular matrices

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$(A - \lambda I) x = 0 \quad \begin{bmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{bmatrix}$$

$$(2 - \lambda)^2 = 0$$

algebraic multiplicity

$$\boxed{\lambda = 2, 2}$$

geometric multiplicity

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

non-diagonalizable

Equation $u_{k+1} = Au_k$
Starting with u_0

$$\Rightarrow A = S \Lambda S^{-1}$$

$$u_{k+1} = A(A(A \dots (Au_0))) \dots$$

$$\boxed{\begin{aligned} u_{k+1} &= A^k u_0 \\ &= S \Lambda^k S^{-1} u_0 \end{aligned}}$$

$\swarrow$ n distinct eigenvectors

the strange way $\rightarrow$ [This also works for less eigenvectors.]

$$u_0 = c_1 x_1 + c_2 x_2 \dots \dots + c_n x_n$$

$$\begin{aligned} Au_0 &= c_1 Ax_1 + c_2 Ax_2 \dots \dots c_n Ax_n \\ &= c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 \dots \dots \end{aligned}$$

$$A^k u_0 = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 \dots \dots$$

$$u_k = \Lambda^k S c$$

eg Fibonacci example.

0, 1, 1, 2, ...

$$\Rightarrow F_{k+2} = F_{k+1} + F_k$$

$$\text{Let } u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

And consider $F_{k+1} = F_k$ equation.

$$\Rightarrow u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \quad \begin{matrix} -(1-\lambda)\lambda - 1 = 0 \\ \lambda(1-\lambda) + 1 = 0 \end{matrix}$$

$$\lambda - \lambda^2 + 1 = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2}$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}$$

$$\lambda_2 = \frac{1 - \sqrt{5}}{2}$$

Now, we can see that the fibonacci numbers approximately grow with $\left(\frac{1 + \sqrt{5}}{2}\right)$ aka the golden ratio

Not $\left(\frac{1 - \sqrt{5}}{2}\right)$ because it is less than

1 so for bigger powers $\left(\frac{1 - \sqrt{5}}{2}\right) \rightarrow 0$

$$u_{100} = c_1 \lambda_1^{100} x_1 + c_2 \lambda_2^{100} x_2$$

$\downarrow$ golden ratio $\downarrow$ ≈ 0

$$u_{100} \approx c_1 \left(\frac{1+\sqrt{5}}{2} \right)^{100} x_1$$

$$\# \begin{bmatrix} 1 - \frac{(1-\sqrt{5})}{2} & 1 \\ 1 & \frac{\sqrt{5}-1}{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} (1+\sqrt{5})/2 & 1 \\ 1 & (\sqrt{5}-1)/2 \end{bmatrix} \begin{pmatrix} 2/\sqrt{5}+1 \end{pmatrix}$$

$$\Rightarrow \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} \lambda \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

$$u_0 = \begin{bmatrix} f_1 \\ f_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Eigenvalues tells us how fast a first order system grows.

→ Difference equations!